

- What is OOP (Object Oriented Programming)?
It is a programming approach (paradigm) where a program is designed using objects that represent real world entities. OOP is based on four main principles:
Encapsulation (Bundling data and methods together, hiding internal details),
Abstraction (Hiding complex implementation details),
Inheritance (reusing features of existing classes),
Polymorphism (allowing one function to behave differently).
- Differences between OOP and Procedural programming.

OOP	Procedural
Program is a collection of interacting object.	Program is a sequence of procedures or function.
Data and methods are encapsulated within objects as a single unit.	Data and functions exist separately, with functions operating on passed data
Data is hidden inside objects and accessed through controlled methods.	Data is often global or shared between multiple functions
Code reuse happens through inheritance and polymorphism.	Code reuse happens through functions and modules
Examples: Python, Java, C++, C#	Examples: C, Pascal

- What are the various Data Types and Data Structures used in python?

The various Data Types in python are:

int - Integer numbers
(e.g., 42, -17, 0)
float - Floating-point numbers
(e.g., 3.14, -0.5)
complex - Complex numbers
(e.g., 3+4j, 2-5j)
str - Strings/text
(e.g., "Hello", 'Python')
bool - Boolean values
(True, False)

Data Structures used in python are:

list - Ordered, mutable collection
(e.g., [1, 2, 3, "apple"])
tuple - Ordered, immutable collection
(e.g., (1, 2, 3, "apple"))
dict - Key-value pairs
(e.g., {"name": "Sam", "age": 24})
set - Unordered collection of unique elements
(e.g., {1, 2, 3, 4})

- What are conditional statements in Python?

Conditional statements in Python are used to make decisions by executing different blocks of code based on whether a given condition is True or False. It allows programs to control the flow of execution and make programs dynamic and intelligent.

1. if statement- Executes a block of code when the condition is true.
2. if-else statement- Executes one block if the condition is true, otherwise executes another block.
3. if-elif-else statement- Used to check multiple conditions.
4. Nested if statement- An if statement inside another if statement.

- What are loops in Python?

Loops in Python are used to repeat a block of code multiple times until a specific condition is met or for a fixed number of iterations. They are essential for automation, repetition, and efficient execution of tasks in Python programs.

1. for loop- Used to iterate over a sequence such as a list, tuple, string, or range.
2. while loop- Executes a block of code as long as the condition is True.

- What are functions in Python and why are they used?

A function in Python is a block of reusable code that performs a specific task.

Functions are defined using the '**def**' keyword and are executed when they are called. It helps in organizing and reusing code efficiently.

- What are the differences between a class and an object

The differences between a class and an object are:

Class

- A class is a blueprint or template for creating objects
- It defines the structure (attributes) and behavior (methods) that objects will have
- It's a logical entity that exists in code as a definition
- Written once and can be used to create multiple objects
- Does not occupy memory until an object is created from it
- Declared using the `class` keyword in Python
- Example: A blueprint of a house showing rooms and layout

Object

- An object is a specific instance created from a class
- It's a physical entity that exists in memory with actual data
- Each object has its own unique set of attribute values
- Multiple objects can be created from a single class
- Occupies memory space when created
- Created using the class constructor
- Example: An actual house built from the blueprint